Alternative formulas for computing $E(X)$.

$\boxed{\text{III}}$ For discrete RV's with

$$\operatorname{supp}(X) = \{0, 1, 2, \dots\}$$

$$E(X) = \sum_{x=0}^{\infty} x \cdot P_X(x)$$

$$= 0 \cdot P_X(0) + \sum_{x=1}^{\infty} x \cdot P_X(x)$$

$2\,P_X(2) \rightarrow$

$3 \cdot P_X(3) \rightarrow$

$$= \boxed{\begin{array}{c} 1 \cdot P_X(1) \\ P_X(2) \\ P_X(3) \\ \vdots \end{array}} \begin{array}{c} \\ + \\ + \\ \\ \end{array} \boxed{\begin{array}{c} P_X(2) \\ P_X(3) \\ \vdots \end{array}} \begin{array}{c} \\ + \\ \\ \end{array} \boxed{\begin{array}{c} P_X(3) \\ \vdots \end{array}}$$

$$P(X>0) + P(X>1) + P(X>2) \cdots$$

$$\therefore \quad E(X) = \sum_{x=0}^{\infty} S_X(x),$$

where $S_X(x) = P(X > x)$ is the survival function.

$\boxed{\text{III}}$ For continuous RV's with $\text{supp}(X) = (0, \infty)$, then

$$E(X) = \int_0^{\infty} S_X(x)\, dx$$

- $$\sum_{x=0}^{n} x = \frac{(0+n)\cdot(n+1)}{2}$$
- $$\sum_{x=0}^{n} a^x = \frac{1 - a^{n+1}}{1 - a}$$ ← # of terms

Ex. Keep rolling a 6-sided die until the first "6" is obtained.

Let X be the # of rolls needed.

Find $E(X)$.

Approach 1. $E(X) = \sum_{\forall x} x \cdot P_X(x)$

$$P_X(2) = P(X=2) = P(\{1\text{st roll} \neq 6\} \cap \{2\text{nd roll} = 6\})$$

$$= \frac{5}{6} \cdot \frac{1}{6}$$

$$P_X(x) = P(\{\text{first } (x-1) \text{ rolls} \neq 6\} \cap \{x\text{th roll} = 6\})$$

$$= \left(\frac{5}{6}\right)^{x-1} \cdot \frac{1}{6}, \quad x = 1, 2, \cdots$$

$$E(X) = \sum_{x=1}^{\infty} x \left(\frac{5}{6}\right)^{x-1} \cdot \frac{1}{6}$$

$$= \frac{1}{6} \sum_{x=1}^{\infty} \frac{d}{dz} g(z) \Big|_{z=\frac{5}{6}}$$

where $g(z) = (z)^{x}$, thus $\frac{d}{dz} g(z) = x \cdot z^{x-1}$

$$= \frac{1}{6} \frac{d}{dz} \sum_{x=1}^{\infty} z^{x} \Big|_{z=\frac{5}{6}}$$

$$= \frac{1}{6} \frac{d}{dz} \sum_{x=0}^{\infty} z^{x+1} \Big|_{z=\frac{5}{6}}$$

$$= \frac{1}{6} \frac{d}{dz} z \cdot \frac{1 - \cancelto{0}{z^{\infty}}}{1-z} \Big|_{z=\frac{5}{6}}$$

$$= \frac{1}{6} \frac{(1-z) + z}{(1-z)^2} \Big|_{z=\frac{5}{6}} = 6.$$

Approach 2. $E(X) = \sum_{x=0}^{\infty} S_X(x)$

$S_X(2) = P(X > 2)$

$= P($ needing more than 2 rolls to obtain the first "6" $)$

$= P($ the 1st and 2nd rolls $\neq 6)$

$= \left(\frac{5}{6}\right)^2$

$S_X(x) = P($ the first x rolls $\neq 6)$

$= \left(\frac{5}{6}\right)^x$

$$E(X) = \sum_{x=0}^{\infty} \left(\frac{5}{6}\right)^{x} = \frac{1-\left(5/6\right)^{\infty}}{1-5/6} = 6.$$

Ex. Let X be a continuous RV with

$$f_X(x) = \begin{cases} p \cdot 2^{p}\, x^{-(p+1)}, & x > 2 \\ 0, & \text{o.w.} \end{cases}$$

where $p > 1$. Find $E(X)$.

Try to use the survival function method.

$$S_X(x) = \int_x^{\infty} p \cdot 2^{p} \cdot x^{-(p+1)}\, dx$$

$$= p \cdot 2^{p} \cdot \frac{1}{-p}\, x^{-p} \Big|_x^{\infty}$$

$$= 2^{p} \cdot x^{-p}, \quad x > 2.$$

NOTE :

since $f_X(x) = 0$ for $x < 2$.

$$E(X) = \begin{cases} \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx = \int_{\boxed{2}}^{\infty} x \cdot p \cdot 2^p \, x^{-(p+1)}\, dx \\ \int_0^{\infty} S_X(x)\, dx \end{cases}$$

$$\int_0^2 1\, dx + \int_2^{\infty} 2^p \cdot x^{-p}\, dx$$

since $S_X(x) = P(X > x)$ for $x \in (0, 2)$.

$$E(X) = \int_0^2 1\, dx + \int_2^{\infty} 2^p \cdot x^{-p}\, dx$$

$$= 2 + 2^p \cdot \frac{1}{-p+1} x^{-p+1} \Big|_2^{\infty}$$

$$= 2 + \frac{2}{p-1}$$